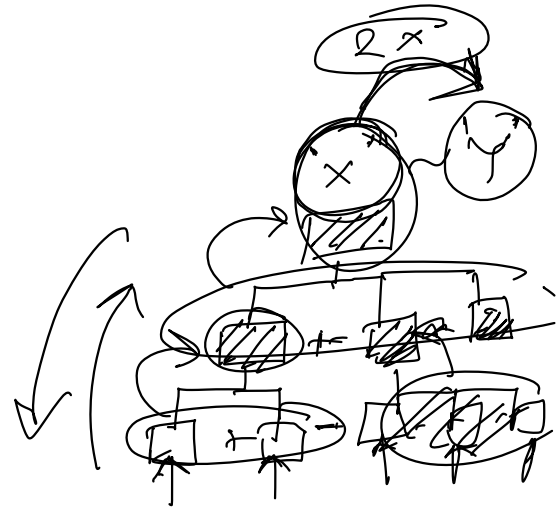


Lec: Guesstimate - 2

Agenda:

- ✓ 1. Problem stmt
- ✓ 2. Assume
- ✓ 3. Basic Terminologies
- ⇒ 4. Approach
 - ✓ 4.1 Clarifying questions
 - ✓ 4.2 Assumptions & Estimate
 - ✓ 4.3 Estimation Tree
 - ✓ 4.4 Bottom up calculation
 - ✓ 4.5 Sanity check



BCG : Boston Consulting Group

Problem Stmt : Estimate AdSense revenue
for the websites.

Flow of Google AdSense:-
(Google / Any business)
1. Advertiser set up Ad

↳ Google Ads

↳ Budget & Target Audience

↳ CPM, CTR, CPC, PPC

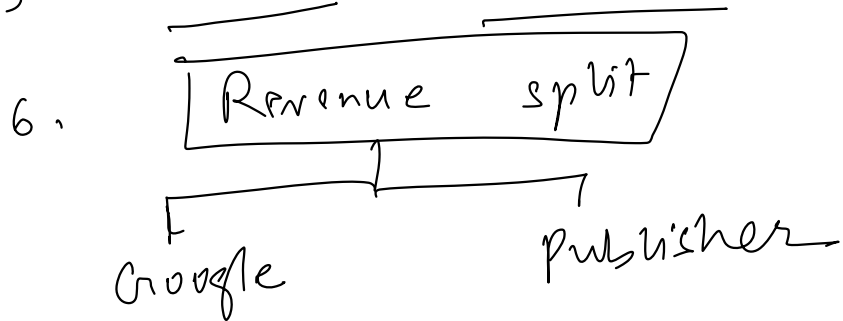


2. Publisher joins the AdSense

3. Google (Algorithm) match the
Ads to the content

4. Ads appear on the publisher
website

5. User sees/click the Ads.



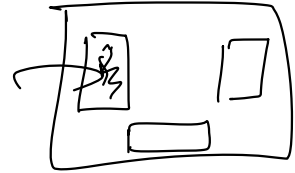
7. Publisher ^{gets} paid.

Basic Terminologies

(1) Click through rate

$$CTR = \frac{\text{Total Clicks}}{\text{Total Impressions}}$$

Click Ads



(2) CPC or PPC

$$= \frac{\text{Advertising cost}}{\text{No. of clicks}}$$

CTR * CPC

→ pay per click

→ Cost per click

③

$$CPM = \left(\frac{\text{Ad Spend}}{\text{Impressions}} \right) * 1000$$

↓
mille
(Thousand) impressions,

Impression
Ads

④ Traffic Acquisition Cost (TAC):

Google retains 32% of revenue

$$\text{Google revenue} = \text{total Ad Revenue} * 0.32$$

$$\underline{\underline{TAC}} = \text{Total Ad Revenue} * \underline{\underline{0.68}}$$

OR

$$TAC = \text{Total Ad Revenue} - \left(\begin{array}{l} \text{Total Ad} \\ \text{Revenue} \\ * 0.32 \end{array} \right)$$

⑤ 80/20 Rule (Pareto Principle) !



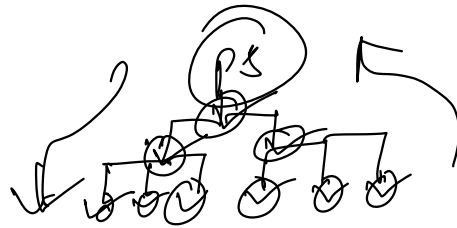
Structured Approach :

1. Ask clarifying questions.

2. High level :

3. State assumptions :

4. Estimation Tree :



- ↓
5. Bottom up !
 6. Sanity check
-

Ad revenue = $\frac{\text{No. of publishers} \times \text{Revenue/day} \times 365 \text{ days}}{\text{Total Revenue}}$

→ Top 20% publishers → 80% Revenue

→ CPC, CTR, CPM

80% of Google

AdSense
Revenue per
year

\$ 2.9 B $\approx$ \$ 2920M

20% of 100
= 20

$\approx$ \$ 0.4 M

Ad Revenue
per day

365
Days.

Top 20%
of publisher
 $\approx$ 20

Impression Ad
Revenue per
day

\$ 0.3M

Click Ad
Revenue
per day

\$ 0.13M

+

Click Ad Revenue
per day

\$ 0.3M

*4
20M

5M

of visitors
per day

25

of click Ads
viewed by
the visitors
per day

0.5%

CTR

\$0.5

CPC

10

of page
visited

*

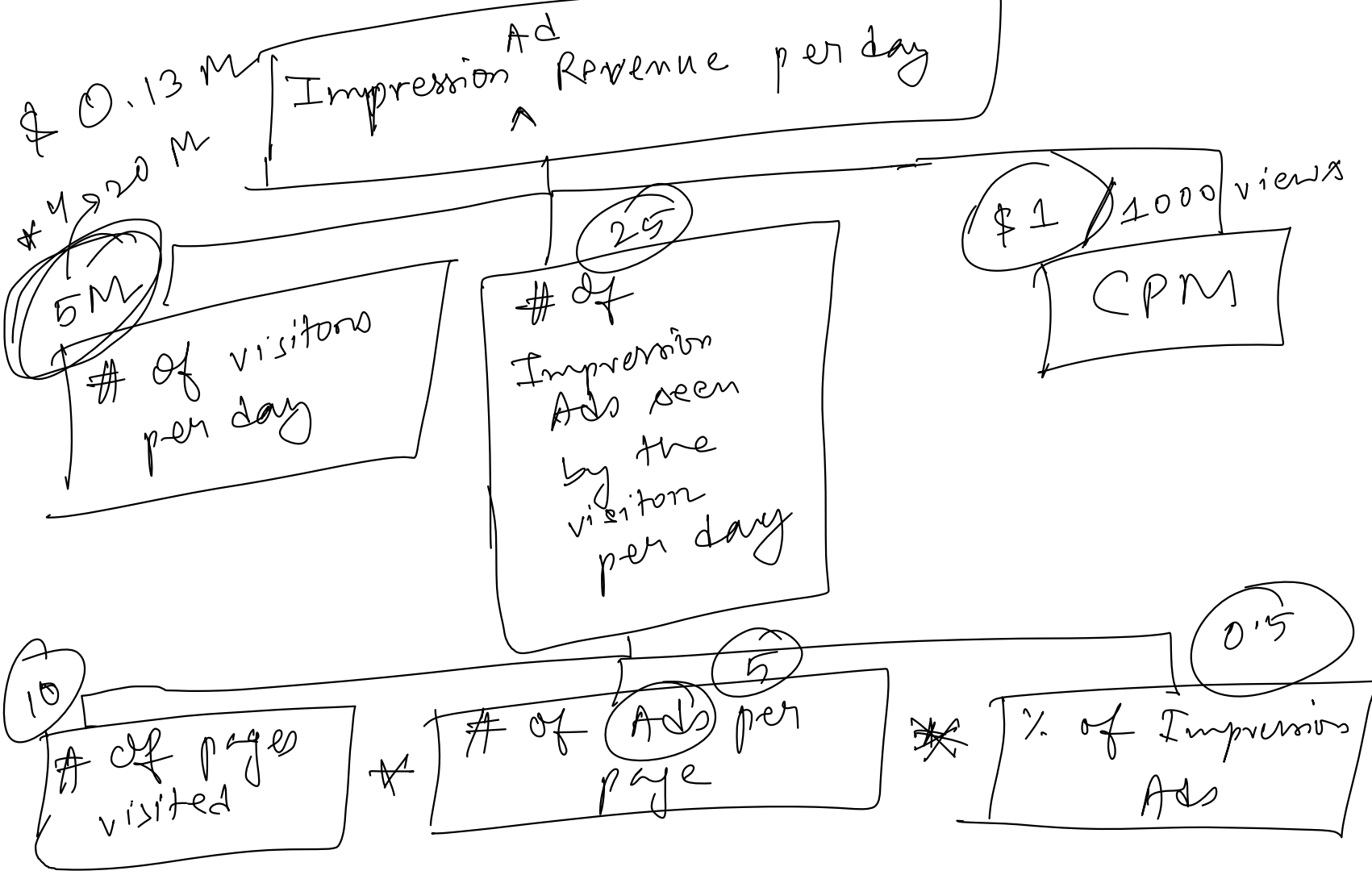
5

of Ads
per page

*

0.5

% of Click
Ads



$$\text{Total Revenue} = \frac{\$2.9B}{0.8} = 3.6B$$

$$0.8 \longrightarrow \$2.9B$$

$$1 \longrightarrow \frac{\$2.9B}{0.8} = \underline{\underline{\$3.6B}} \leftarrow \underline{\underline{\text{Answer}}}$$

Sanity Check:

Actual Revenue =

\$ 17B

